In the table below we give some functions, their derivatives and their differentials side by side for comparative purpose.

| S. No. | Function | Derivative | Differential |
|---|---|---|---|
| 1 | $f(x) = x^n$ | $f'(x) = nx^{n-1}$ | $df = nx^{n-1}dx$ |
| 2 | $f(x) = \cos(x^2 + 7x)$ | $f'(x) = -\sin(x^2 + 7x)(2x + 7)$ | $df = -\sin(x^2 + 7x)(2x + 7)dx$ |
| 3 | $f(x) = \cot(x^2)$ | $f'(x) = -\operatorname{cosec}^2(x^2)2x$ | $df = -\operatorname{cosec}^2(x^2)2x\,dx$ |
| 4 | $f(x) = \sin^{-1}(x)$ | $f'(x) = \frac{1}{\sqrt{1-x^2}}$ | $df = \frac{1}{\sqrt{1-x^2}}dx$ |
| 5 | $f(x) = \tan^{-1} x$ | $f'(x) = \frac{1}{1+x^2}$ | $df = \frac{1}{1+x^2}dx$ |
| 6 | $f(x) = e^{x^3-5x+7}$ | $f'(x) = e^{x^3-5x+7}(3x^2 - 5)$ | $df = e^{x^3-5x+7}(3x^2 - 5)dx$ |
| 7 | $f(x) = \log(x^2 + 1)$ | $f'(x) = \frac{2x}{x^2+1}$ | $df = \frac{2x}{x^2+1}dx$ |

Next we look at the properties of differentials. These results easily follow from the definition of differential and the rules for differentiation. We give a proof for (5) below and the other proofs are left as exercises.

## Properties of Differentials

Here we consider real-valued functions of real variable.

(1) If $f$ is a constant function, then $df = 0$.

(2) If $f(x) = x$ identity function, then $df = 1dx$.

(3) If $f$ is differentiable and $c \in \mathbb{R}$, then $d(cf) = cf'(x)dx$.

(4) If $f, g$ are differentiable, then $d(f + g) = df + dg = f'(x)dx + g'(x)dx$.

(5) If $f, g$ are differentiable, then $d(fg) = fdg + gdf = (f(x)g'(x) + f'(x)g(x))dx$.

(6) If $f, g$ are differentiable, then $d(f/g) = \frac{gdf - fdg}{g^2} = \frac{g(x)f'(x) - f(x)g'(x)}{g^2(x)}dx$, where $g(x) \neq 0$.

(7) If $f, g$ are differentiable and $h = f \circ g$ is defined, then $dh = f'(g(x))g'(x)dx$.

(8) If $h(x) = e^{f(x)}$, then $dh = e^{f(x)}f'(x)dx$.

(9) If $f(x) > 0$ for all $x$ and $g(x) = \log(f(x))$, then $dg = \frac{f'(x)}{f(x)}dx$.



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